Lifting the Exponent Lemma — Solutions

Problem 1. (Romania TST 1994 Test 1 Problem 2) Let n be an odd positive integer. Prove that $((n-1)^n+1)^2$ divides $n(n-1)^{(n-1)^n+1}+n$.

Solution. Take any prime divisor p of $((n-1)^n+1)^2$. We must have $p \mid (n-1)^n+1$.

As $n=(n-1)+1 \mid (n-1)^n+1$, we have

$$\begin{aligned} v_p(n(n-1)^{(n-1)^n+1}+n) &= v_p((n-1)^{(n-1)^n+1}+1)+v_p(n) \\ &= v_p((n-1)^n+1)+v_p\left(\frac{(n-1)^n+1}{n}\right)+v_p(n) \\ &= v_p((n-1)^n+1)+v_p((n-1)^n+1) \\ &= 2v_p((n-1)^n+1) \\ &= v_p(((n-1)^n+1)^2) \end{aligned}$$

by the lifting the exponent lemma. Since this holds for all primes p dividing $((n-1)^n+1)^2$, we must have $((n-1)^n+1)^2 \mid n(n-1)^{(n-1)^n+1}+n$. $\square$

Problem 2. (North Macedonia TST for Balkan MO Test 2 Problem 2) Given a prime p and a positive integer a, let q be a prime divisor of $\dfrac{a^{p^3}-1}{a^{p^2}-1}$ and $q \neq p$. Prove that $q \equiv 1 \pmod{p^3}$.

Solution. If $q \mid a^{p^2}-1$, then

$$v_q(a^{p^3}-1)=v_q(a^{p^2}-1)+v_q(p)=v_q(a^{p^2}-1)$$

by the lifting the exponent lemma. This contradicts the assumption that $q \mid \dfrac{a^{p^3}-1}{a^{p^2}-1}$. Therefore, we must have $a^{p^2} \not\equiv 1 \pmod{q}$. As $a^{p^3} \equiv 1 \pmod{q}$, the order d of a modulo q must divide p^3. But since $d \nmid p^2$, the only possibility is $d=p^3$. By the Fermat little theorem, we have $a^{q-1} \equiv 1 \pmod{q}$, and hence $d \mid q-1$. This yields $q \equiv 1 \pmod{p^3}$. $\square$

Problem 3. Let a and b be distinct real numbers such that a^n-b^n is an integer for every positive integer n. Prove that a and b are integers.

Solution. Since both $a-b$ and $a+b=\dfrac{a^2-b^2}{a-b}$ are rational, so are a and b. Let d be the smallest positive integer such that $a_1=da \in \mathbb{Z}$ and $b_1=db \in \mathbb{Z}$.

If $d \geqslant 2$, there exists a prime p dividing d. Note that $p \nmid (a_1, b_1)$ since otherwise we can replace d by $\dfrac{d}{p}$, contradicting the minimality of d. As $a^n - b^n \in \mathbb{Z}$, we have $d^n \mid a_1^n - b_1^n$, and hence $p^n \mid a_1^n - b_1^n$. If $p \geqslant 3$, by the lifting the exponent lemma, we have

$$v_p(a_1^n - b_1^n) = v_p(a_1 - b_1) + v_p(n).$$

This should be at least n since $p^n \mid a_1^n - b_1^n$. However, if $p \nmid n$, then $v_p(a_1 - b_1) + v_p(n) = v_p(a_1 - b_1)$ is fixed, hence it cannot be at least n if n is sufficiently large. Similarly, if $p = 2$, we choose $n = 2k$ where k is odd. Then we have

$$v_2(a_1^n - b_1^n) = v_2(a_1^2 - b_1^2) + v_2(n) - 1 = v_2(a_1^2 - b_1^2).$$

This is fixed, so it cannot be at least n for sufficiently large n. This again yields a contradiction.

Therefore, we must have $d = 1$, which means a and b are integers. □

Problem 4. Find all nonnegative integers m, n and all prime numbers p satisfying

$$2p^m - n^p = 1.$$

Solution. Answer: $m = 0$, $n = 1$ and p can be any prime.

If $n = 1$, the equation becomes $p^m = 1$. This holds iff $m = 0$ (where p is arbitrary). Similarly, if $m = 0$, we must have $n = 1$. Now assume $n \geqslant 2$ and $m \geqslant 1$.

If $p = 2$, the equation gives $2^{m+1} = n^2 + 1$. Note that n must be odd. So $n^2 + 1 \equiv 2 \pmod 8$, contradicting $4 \mid 2^{m+1}$.

If $p \geqslant 3$, we have $n + 1 \mid n^p + 1 = 2p^m$. Note that n must be odd. So we have $\dfrac{n+1}{2} \mid p^m$. Let $\dfrac{n+1}{2} = p^k$, i.e. $n = 2p^k - 1$. By the lifting the exponent lemma, we have

$$v_p(n^p + 1) = v_p(n + 1) + v_p(p) = k + 1.$$

This implies $m = k + 1$. Thus,

$$p = \frac{2p^{k+1}}{2p^k} = \frac{n^p + 1}{n + 1} \geqslant \frac{n^3 + 1}{n + 1} = n^2 - n + 1 \geqslant (2p - 1)^2 - (2p - 1) + 1$$

since $n = 2p^k - 1 \geqslant 2p - 1$ and $n^2 - n + 1$ is increasing for $n \geqslant 1$. One easily checks that this is impossible as $p \geqslant 3$. □

Problem 5. (IMO 1999 Problem 4 modified) Find all the pairs of positive integers (x, p) such that p is a prime and x^{p-1} is a divisor of $(p-1)^x + 1$.

Solution. Answer: $(p, x) = (p, 1), (2, 2), (3, 3)$ where p can be any prime.

If $p = 2$, we need $x \mid 2$, which holds iff $x = 1, 2$.

If $p \geqslant 3$, then $(p-1)^x+1$ is odd, and so x is odd. If $x = 1$, then p can be any prime. If $x > 1$, let $q \geqslant 3$ be the smallest prime divisor of x. Let d be the order of $p-1$ modulo q. Since $q \mid x^{p-1} \mid (p-1)^x+1$, we have $(p-1)^x \equiv -1 \pmod q$. Therefore, $d \nmid x$ but $d \mid 2x$. By the Fermat little theorem, we have $d \mid q-1$. Thus, $d \mid (2x, q-1) = 2$ (note that $(x, q-1) = 1$ since q is the smallest prime divisor of x). As $d \nmid x$, we must have $d = 2$. Thus, $(p-1)^2 \equiv 1 \pmod q$, and hence $p - 1 \equiv -1 \pmod q$. It follows that $q \mid p$, so $q = p$.

As $x^{p-1} \mid (p-1)^x + 1$, we have

$$(p-1)v_p(x) = v_p(x^{p-1}) \leqslant v_p((p-1)^x + 1) = v_p((p-1) + 1) + v_p(x) = 1 + v_p(x)$$

by the lifting the exponent lemma. This yields $(p-2)v_p(x) \leqslant 1$, and hence $p = 3$ and $v_3(x) = 1$. If $x = 3$, we check that $3^2 \mid 2^3 + 1$, so $(p, x) = (3, 3)$ is a solution. If $x > 3$, let r be the smallest prime divisor of x other than 3, and let f be the order of $3 - 1 = 2$ modulo r. As above, we must have $f \nmid x$ and $f \mid (2x, r-1) \mid 6$ since r is the smallest prime divisor of x other than 3. From this, it follows that $2^6 \equiv 1 \pmod r$. The only possibility is $r = 7$. But then $f = 3$, which contradicts $f \nmid x$.

To conclude, the only solutions are $(p, x) = (p, 1), (2, 2), (3, 3)$ where p can be any prime. $\square$

Problem 6. Solve the equation $(a^5 + b)(b^5 + a) = 2^c$ in positive integers.

Solution. Answer: $(a, b, c) = (1, 1, 2)$.

Let $a^5 + b = 2^k$ and $b^5 + a = 2^\ell$. If both a and b are even, then $v_2(a^5) = v_2(b)$ and $v_2(b^5) = v_2(a)$. This yields $5v_2(a) = v_2(b)$ and $5v_2(b) = v_2(a)$, which is impossible. As a and b have the same parity, they must be odd. WLOG assume $k \leqslant \ell$.

From $a^5 + b \equiv 0 \pmod{2^k}$ and $b^5 + a \equiv 0 \pmod{2^k}$, we have $a^{25} \equiv -b^5 \equiv a \pmod{2^k}$. This implies $2^k \mid a^{24} - 1$ as a is odd. By the lifting the exponent lemma, we have

$$v_2(a^{24} - 1) = v_2(a^2 - 1) + v_2(24) - 1 = v_2(a^2 - 1) + 2.$$

It follows that $k \leqslant v_2(a^2 - 1) + 2$, and so $a^2 - 1 \geqslant 2^{k-2}$. Thus,

$$a^5 + b = 2^k \leqslant 4(a^2 - 1).$$

If $a \geqslant 2$, this implies

$$a^5 + b \leqslant 4(a^2 - 1) \leqslant 2(a^4 - 1) \leqslant a(a^4 - 1) = a^5 - a,$$

contradiction. Therefore, we must have $a = 1$. The equations become $1 + b = 2^k$ and $b^5 + 1 = 2^\ell$. As $v_2(b^5 + 1) = v_2(b + 1)$, we need $k = \ell$. This yields $1 + b = b^5 + 1$, so that $b = 1$. One checks that $a = b = 1$ is a solution, where $k = \ell = 1$ and $c = 2$. $\square$

Problem 7. (China TST 2002 Quiz 7 Problem 2) Find all nonnegative integers m and n, such that $(2^n - 1) \cdot (3^n - 1) = m^2$.

Solution. Answer: $m = n = 0$.

If $m = 0$ or $n = 0$, it is easy to check that $m = n = 0$, which is a solution. Now, assume $m, n \geqslant 1$. If n is odd, then

$$v_2((2^n - 1)(3^n - 1)) = v_2(3^n - 1) = v_2(3 - 1) = 1.$$

Then $(2^n - 1)(3^n - 1)$ cannot be a perfect square. Let $n = 2a$. By the lifting the exponent lemma, we have

$$v_3((2^n - 1)(3^n - 1)) = v_3(2^n - 1) = v_3(2^2 - 1) + v_3(a) = 1 + v_3(a).$$

Since this must be even, we have $v_3(a) \geqslant 1$. Let $a = 3b$, which means $n = 6b$. Note that

$$(2^n - 1)(3^n - 1) = (64^b - 1)(729^b - 1) \equiv (2^b - 1)(16^b - 1) = (2^b - 1)^2(2^b + 1)(4^b + 1) \pmod{31}.$$

We claim that $5 \mid b$. If not, we must have $2^b - 1 \not\equiv 0 \pmod{31}$ as the order of 2 modulo 31 is 5. This shows $(2^b + 1)(4^b + 1)$ is a quadratic residue modulo 31. When $b \equiv 1, 2, 3, 4 \pmod{5}$, we have $(2^b + 1)(4^b + 1) \equiv 3 \cdot 5, 5 \cdot 17, 9 \cdot 3, 17 \cdot 9 \pmod{31}$ respectively. One checks that

$$\begin{aligned}
&\left(\frac{3}{31}\right) = -\left(\frac{31}{3}\right) = -\left(\frac{1}{3}\right) = -1,\\
&\left(\frac{5}{31}\right) = \left(\frac{31}{5}\right) = \left(\frac{1}{5}\right) = 1,\\
&\left(\frac{17}{31}\right) = \left(\frac{31}{17}\right) = \left(\frac{-3}{17}\right) = \left(\frac{3}{17}\right) = \left(\frac{17}{3}\right) = \left(\frac{2}{3}\right) = -1.
\end{aligned}$$

Therefore, 3 and 17 are quadratic nonresidues modulo 31, while 5 is a quadratic residue modulo 31. It follows that $3 \cdot 5, 5 \cdot 17, 9 \cdot 3, 17 \cdot 9$ are all quadratic nonresidues modulo 31. This is a contradiction.

Therefore, we must have $5 \mid b$. Let $b = 5c$, which means $n = 30c$. By the lifting the exponent lemma, we have

$$\begin{aligned}
v_{11}((2^n - 1)(3^n - 1)) &= v_{11}(2^{30c} - 1) + v_{11}(3^{30c} - 1)\\
&= v_{11}(2^{10} - 1) + v_{11}(3c) + v_{11}(3^{10} - 1) + v_{11}(3c)\\
&= v_{11}(1023) + v_{11}(59048) + 2v_{11}(3c)\\
&= 1 + 2 + 2v_{11}(3c).
\end{aligned}$$

This shows $(2^n - 1)(3^n - 1)$ cannot be a perfect square as the exponent of 11 is odd. $\square$